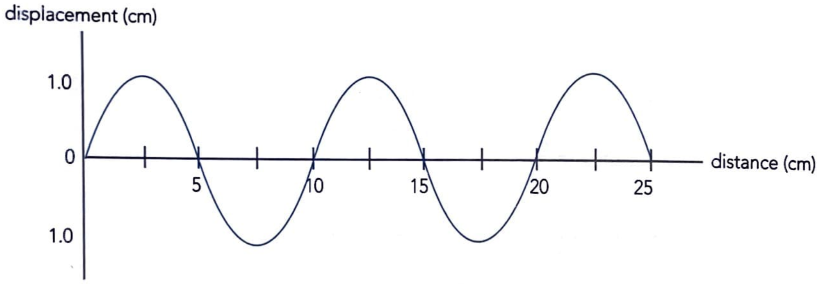
REVIEW QUESTIONS

Chapter 5: Waves

1. A typical value for the speed of sound is 340 ms^{-1}. Use this value to calculate the wavelength corresponding to the following sound frequencies:
   (a) A school siren with a frequency of 500 Hz.
   (b) The frequency of sound that humans can most easily detect (about 3000 Hz).
   (c) A middle C note played on a piano (264 Hz).
   (d) The sound from a dog whistle (15 kHz).
   (e) An ultrasound scanner operating at 3.5 MHz.

2. Julian and Brook, two keen surfers, are on a beach watching the waves coming in to the shore. They estimate the crests to be 20 m apart and note that 12 waves reach the shore each minute.
   (a) Determine:
      (i) the frequency of the waves,
      (ii) the period of the waves,
      (iii) the velocity of the waves.
   (b) If the surfers could successfully ride one of these waves from 200 m offshore how long would their surf ride last? (Assume that the waves are travelling directly to shore.)

3.

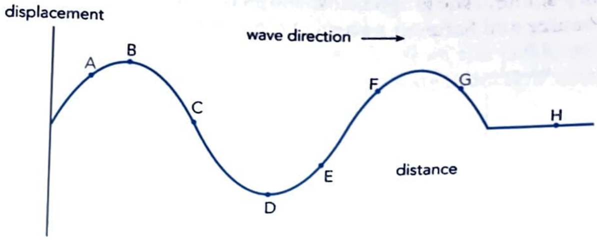


A ripple tank is used to generate water waves as shown in the displacement distance graph above. Determine:
   (a) the wavelength of the waves,
   (b) the amplitude of the waves,
   (c) the velocity of the waves given that the frequency is 3.25 Hz.

4. A ball thrown in the middle of a swimming pool causes a series of ripples in all directions. In an experiment it was noted that the ripples reached the edge of the pool (2.40 m away) in 6.50 s and that a total of 18 ripples had been formed in this time. For these ripples determine their:
   (a) period,
   (b) wavelength,
   (c) velocity.

5. During a practice session for a 100 m race, Paula decided to time her friend Beth by standing at the finish line with a stopwatch and getting another friend to fire a starting pistol at the start line. She started the stopwatch as soon as she heard the sound of the starting pistol and stopped it as Beth went over the finish line.
   (a) This method is likely to lead to timing errors. Explain why.
   (b) Paula timed Beth at 14.2 s for the 100 m sprint. What is the more likely correct time for this sprint? (Assume velocity of sound to be 344 ms^{-1}.)

6.



The motion of a water wave in a large ripple tank is illustrated above. Assume that at the instant shown, the wave is moving to the right and small corks are floating at points A to G. Which cork (or corks) shown:

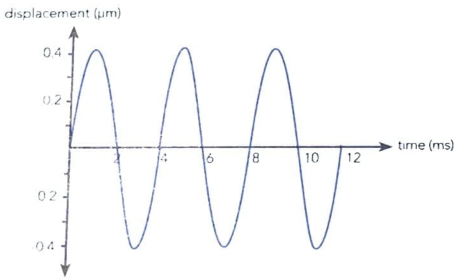
(a) are moving the fastest? ________

(b) are stationary? ________

(c) are moving vertically down? ________

(d) are moving in phase with each other? ________

7. The graph below represents the displacement from mean position of a molecule of air (X) as a sound wave passes by.



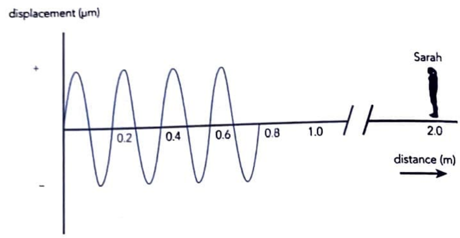
(a) Determine the period and frequency of this wave.

(b) The nearest air molecule moving in phase with molecule X is 1.40 m away. What is the speed of this sound wave?

(c) What is the maximum amplitude of the air molecule X?

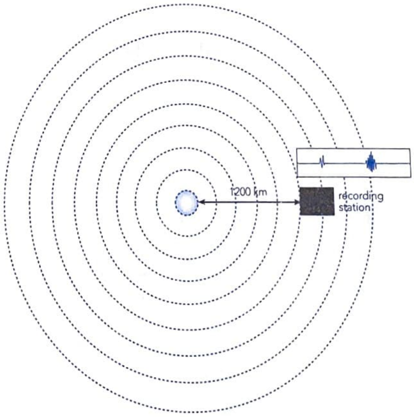
(d) Select a point in time from the graph during which air molecule X would have maximum velocity.

8. Sarah is standing 2.0 m away from a loud speaker which is emitting a single frequency sound. The graph below shows the air pressure variation between the sound source and Sarah at a particular instant in time (t = 0).



(a) Assuming the speed of sound is 340 ms^{-1}, determine the frequency and wavelength of this sound.
(b) If the graph shows the sound at t = 0, how long before Sarah hears the sound?
(c) Sketch an air pressure versus time graph for t = 0 to t = 5 ms for the position where Sarah is standing.

9. An seismic recording station is a distance of 1200 km from where an earthquake has just occured. It detects and records the arrival of both P and S waves.



(a) Assuming an average velocity of 5.2 km/sec and 3.3 km/sec for the P and S waves calculate the time difference of arrival of the two waves.
(b) A more direct method of calculating this distance is to use a conversion factor established from the study of many previous seismic events. Typically it is 9.0 km/sec. Use this value to determine the distance to the earthquake location and comment on any differences.
(c) Suggest how the actual location of the earthquake can be established. Explain your method clearly.

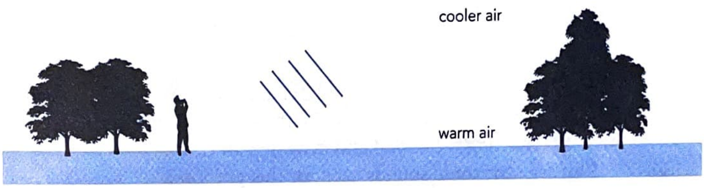
10. Maxwell is investigating how the intensity of the sound produced by an alarm bell varies with distance. When he is 5.0 m from the bell he finds the intensity to be 2.85×10^{-2} Wm^{-2}. If we assume no effects from sound reflections or absorption determine the likely intensities that Maxwell will find at distances of:
(a) 10.0 m from the bell
(b) 20.0 m from the bell.

11. An observer standing 12.0 m from an operating jackhammer measures the sound intensity at that point and finds it to be 1.0×10^{-3} Wm^{-2}. She also finds that by wearing a set of high quality earmuffs the intensity can reduced by a factor of 100. Determine, for comparison, how far from the jackhammer she would need to stand to get the same amount of intensity reduction without the use of earmuffs.

Reflection, Refraction, Diffraction of Waves

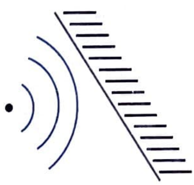
12. When sound waves enter different media, their speed, wavelength and direction can change. However frequency never changes. Explain why it is not possible for frequency to change.

13. On a warm, calm day the air near ground level is warmer than that further above. Complete the diagram below to show how sound is affected by the different layers of air as it travels away from a student shouting on an oval.

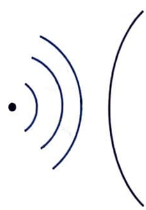


14. Complete the following diagrams to show how the sound waves are reflected.

(a) oblique flat surface



(b) convex surface



15. A sound wave travels from air into water. How are the following likely to be affected?

(i) speed
(ii) wavelength
(iii) frequency
(iv) direction
(v) intensity

16. Complete the following to show how diffraction varies with wavelength and magnitude of opening.

(i)

(ii)

(iii)

17. A pop concert fan notices that he is able to distinguish the high frequency sounds of the music much better if directly in front of the band rather than at the side. Explain clearly how this is possible.

18. Dolphins can locate objects by emitting sounds of up to 150 kHz in short pulses.
   (a) Assuming the velocity of sound in sea water is to be 1530 ms^{-1} determine the wavelength of these sounds.
   (b) Why are the higher frequency sounds likely to be more effective in the location of objects by dolphins? Give two reasons.
   (c) What would be the minimum sized object the dolphins could easily detect by this method? Explain.

19. In our homes, we can readily hear sounds coming from nearby rooms even though we may not actually see the sound source.
   (a) Explain how this is possible.
   (b) Which type of sounds are we likely to hear best?

20. A partly open door has a gap of 15 cm through which sound can diffract. Which range of sound frequencies will diffract best through this gap?

21. Modern cameras are often fitted with ultrasonic focussing systems. These determine the distance of the subject from the camera by measuring the time delay between emitting the sound and detecting an echo.
   (a) Why are ultrasonic sounds used for this purpose?
   (b) What are the limitations of this method?
   (c) What would be the time delay for a subject 10.0 m from the camera?

Wave Interaction

22. Complete the following diagrams to show the resultant displacement when the two wave pulses shown coincide totally.

(a)

(b)

23. Three similar tuning forks, X, Y and Z, are sounded together in pairs in order to determine the unknown frequency of one of them. The frequencies of X and Y are 440 Hz and 445 Hz respectively. When Z is sounded with X, 10 beats are heard in 5 seconds. When it is sounded with Y, 15 beats are heard in 5 seconds. What is the frequency of tuning fork Z?

24. Beats and standing waves are each caused by the interaction of two separate wave motions occurring in the same medium. Describe the similarities and differences between the two wave motions causing each of these effects.

25. Two speakers, A and B, are emitting sound waves in the same direction. The wave-lengths are 1.40 m and 1.50 m respectively. Assume the velocity of sound is 340 ms^{-1}.

(a) What will be the distance between successive in phase positions?

(b) What will be the time difference between the occurrence of maximum sound intensities at any given point?

26.

Two speakers, A and B, are emitting sounds of equal frequency and intensity towards each other as shown. A microphone connected to a CRO is used to check the sound intensity level at all points between A and B.

(a) Describe how the sound intensity level may vary between A and B.

(b) What type of wave pattern exists between the two speakers.

(c) What is the name given to points of:

(i) minimum intensity?

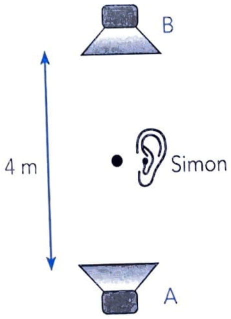
(ii) maximum intensity?

(d) What is the significance of the distance between successive points of equal intensity?

27. Simon connects two speakers (A and B) to a frequency generator and tests the sound he hears as he walks directly between them. The speakers are 4.0 m apart as they are sounded together as Simon walks from A to B.

(a) Is he likely to hear a loud or soft sound when he is midway between the speakers? Explain.

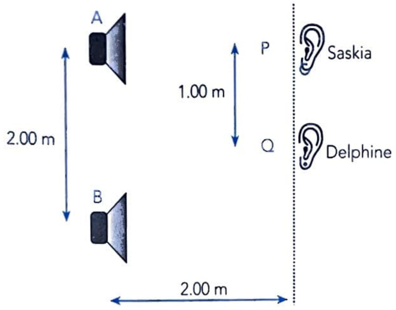
(b) He walks from the midpoint 1.0 m towards B and encounters two points of minimum and two points maximum sound intensity. Determine the wavelength of the sound being used.



28. Chelsea and Hannah connect two speakers (A and B) to a single frequency sound source and then investigate the loudness of the sound produced at different points of the laboratory. The speakers are 2.00 m apart and Hannah stands at a position Q which is exactly midway between the speakers and a perpendicular distance of 2.00 m from them.

(a) If Hannah begins to walk towards Chelsea, in what way will the sound that she hears vary? Explain.

(b) Chelsea, who is standing directly in front of speaker A, as shown, cannot hear any sound at this position. Determine the minimum frequency of the sound being emitted by the two speakers (assume the speed of sound is 340 ms^{-1}).

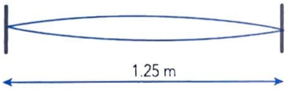


29. Samuel plucks a harp string 1.25 m long so that it is vibrating at its fundamental frequency.

(a) What name is given to this type of wave pattern?

(b) Sketch two other possible modes of vibration for this string.

(c) What is the maximum wavelength that this string can have when vibrating?



30. Rebecca and Pamela set up a resonating air column in the laboratory using an "A" note tuning fork (440 Hz). They then measured the sound intensity levels at different points inside the pipe. The pipe was 1.20 m in length. Beginning at one end of the pipe they found:

(i) maximum intensity at 0.00 m, 0.40 m, and 0.80 m.

(ii) minimum intensity at 0.20 m and 0.60 m.

(a) From the above information only - determine the wavelength of the sound.

(b) Determine the speed of sound in this experiment.

(c) Did the two students use an open or closed pipe as a resonating air column? Explain.